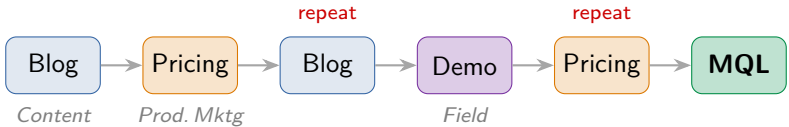
Attribution Modeling

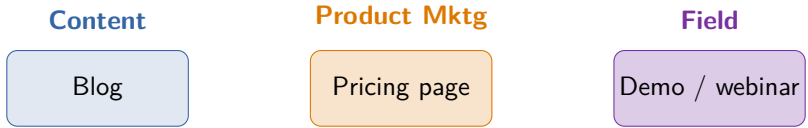
Who gets credit for the conversion? Heuristics · Markov chains · Shapley values · Shapley on ML

One prospect's path to becoming an MQL



5 touchpoints, but only **3 distinct channels** — Blog, Pricing, Demo.

Three teams, three assets

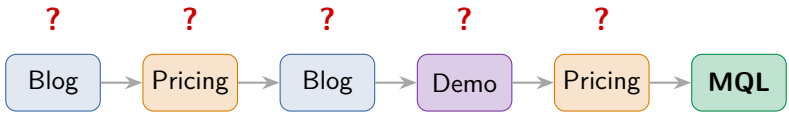


Each team asks the same question:

"Did my asset drive the MQL?"

Budget and headcount ride on the answer.

5 touches, 1 conversion — who gets the credit?



We must split **one** conversion's credit across the channels that led to it.

How much of the MQL does each of Blog, Pricing, Demo deserve?

We see *that* they converted, not *why*

- The data shows the touches and the outcome — not the cause.
- A channel being *present* in winning journeys $\neq$ that channel *caused* the win.
- Popular channels look good simply by appearing everywhere.

So: how do we turn "was present" into "deserves credit"?

Attribution = a share of credit per channel

Goal

Assign each channel a **credit** number, so the shares are comparable — which assets actually move MQLs.

- Bigger credit $\Rightarrow$ that channel was more responsible for conversions.
- Lets each team see the return on its work.

From messy journey to distinct channels

For now, *for simplicity*, we collapse to the *set* of distinct channels ↓

We just threw away the repeats — Blog and Pricing were each hit twice.
Hold that thought.

First touch vs last touch

This journey — which single channel deserves the credit?

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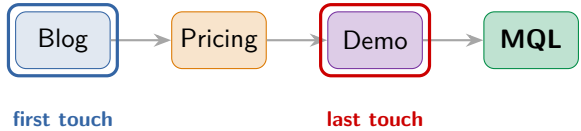
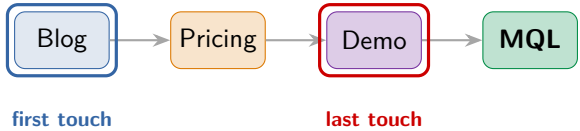


First-touch credits Blog. **Last-touch** credits Demo. **Opposite!**

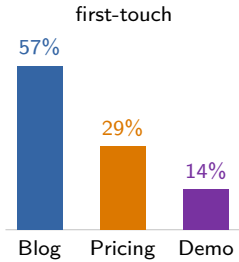
First touch vs last touch

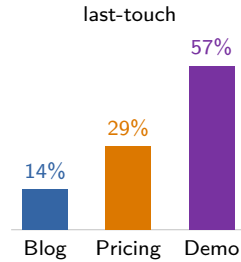
This journey — which single channel deserves the credit?



First-touch credits Blog. **Last-touch** credits Demo. **Opposite!**

Apply each rule across *all* journeys ⇒ channel-level credit:





Each single-touch rule has a built-in bias

First-touch

- Over-credits *top-of-funnel* — awareness, discovery.
- Ignores everything that nurtured and closed.

Last-touch

- Over-credits *the closer* — the final click.
- Ignores what built the interest in the first place.

Either way: you reward one touch and erase the rest.

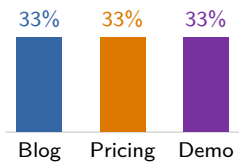
Spreading the credit: multi-touch rules

Give *every* touch some credit — but how much?

(positions on the journey Blog → Pricing → Demo)

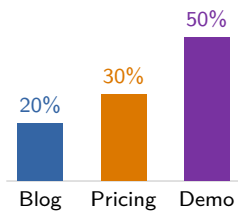
Linear

equal split



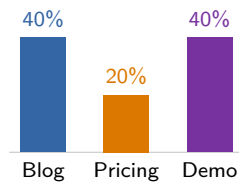
Time-decay

recent weighted



U-shaped

ends weighted



Each is a fixed *position-based* recipe — it never looks at the outcomes.

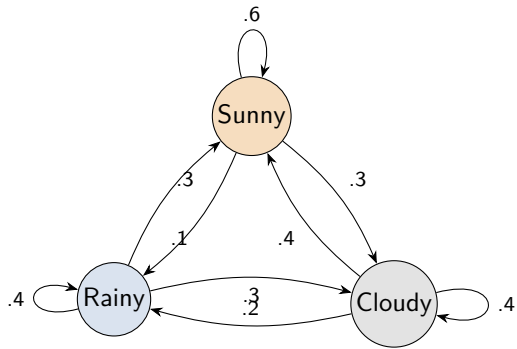
Why 40/40/20, and not 30/30/40?

These splits are **arbitrary** — picked by hand, justified by *nothing in the data*.

- Every rule so far decides credit by *position*, before looking at outcomes.
- Change the recipe, change the "winner" — with no principled way to choose.

Data-driven idea: let the *data* decide — which channels actually move conversions?

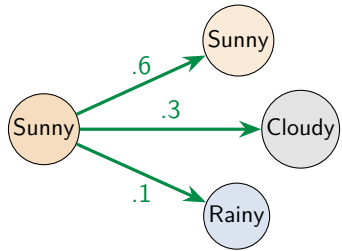
A Markov chain in one picture



States + transition probabilities. Each arrow: "given today, the chance of tomorrow."

From one state, the chances sum to 1

Given **today is Sunny**, tomorrow is exactly one of:



$$\underbrace{0.6}_{\text{Sunny}} + \underbrace{0.3}_{\text{Cloudy}} + \underbrace{0.1}_{\text{Rainy}} = 1$$

From any state, the outgoing arrows cover every option — they **always sum to 1**.

Memoryless: only *now* matters

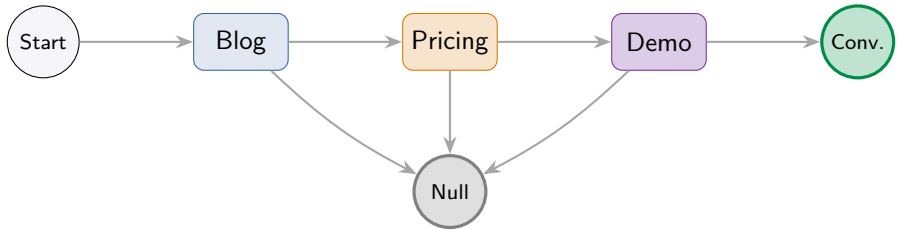
The Markov property

The next state depends **only on the current state** — not on the full history of how you got there.

Weather: tomorrow depends on today, not on what happened last week.

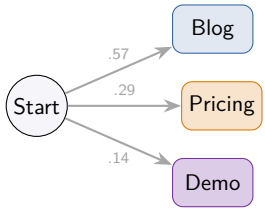
Real journeys *do* have memory — first-order Markov is a deliberate simplification. (Higher-order chains exist; we keep it first-order here.)

A journey is a walk through channel-states



- States = the channels, plus **Start**, **Convert**, and **Null** (no conversion).
- **Convert** and **Null** are *absorbing*: once you arrive, you stay.

Estimate transition probabilities by counting

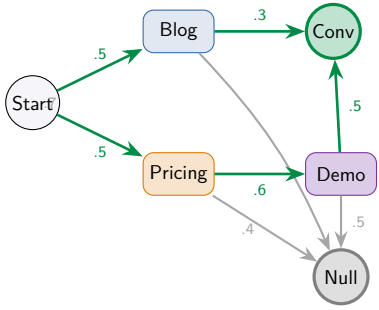


Count transitions; divide by row totals. The **Start** row from our data:

$$\text{Start} \to (\text{B}, \text{P}, \text{D}) = (.57,\ .29,\ .14)$$

Every state's outgoing probabilities sum to 1.

How likely is conversion? Add up the paths



illustrative acyclic graph — green = converting paths

$$P(\text{conv}) = \sum_{\text{Start}\to\cdots\to\text{Conv}} \prod_{\text{edges}} p$$

Two converting paths:

- St→Blog→Conv: $.5 \times .3 = .15$
- St→Pric→Demo→Conv: $.5 \times .6 \times .5 = .15$

$$P(\text{conv}) = .15 + .15 = \mathbf{0.30}$$

With repeat visits (cycles) the paths are infinite — use the *fundamental matrix*. The script does this for the real numbers.

Remove a channel, watch conversion drop

Removal effect = a channel's importance

$$\text{credit}_i \;\propto\; \frac{P(\text{convert}) - P(\text{convert} \mid i \text{ removed})}{P(\text{convert})}$$

- **Remove a channel** = redirect all its incoming transitions to **Null**.
- The bigger the drop in conversion, the more that channel mattered.
- Compute for every channel, then normalize to shares.

Removing Pricing

full chain

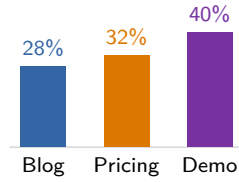
$$P(\text{conv}) = \mathbf{0.386}$$

Pricing $\to$ Null

$$P(\text{conv}) = \mathbf{0.157}$$

$$\text{removal effect}_{\text{Pricing}} = \frac{0.386 - 0.157}{0.386} \approx 0.59 \quad \text{(a 59\% drop)}$$

Repeat for all three, normalize $\Rightarrow$ Markov credit shares:



Markov's question

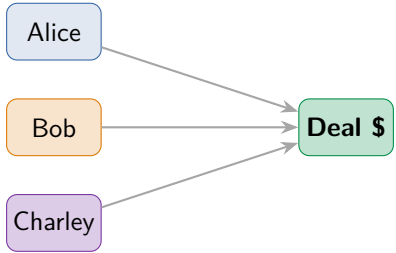
Markov asks:

*"What **breaks** if this channel disappears?"*

A natural "what if we turned it off?" view of importance.

Next: a completely different question — *fairness*.

Three people close one deal — split the bonus fairly



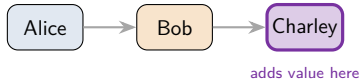
They produced **one** outcome *together*.

What is each person's fair share of the bonus?

Channels are the same: several touches jointly produce **one** conversion.

Average marginal contribution over all join orders

- Imagine the players joining a team *one at a time*.
- A player's **marginal contribution** = how much value they *add* when they join.
- **Shapley value** = that contribution, *averaged over every possible order*.



one ordering: Alice, then Bob, then Charley — measure what Charley adds

All 6 orders — Blog's marginal contribution

$v(S)$ = conversion rate of journeys with exactly set S.

order	set before Blog	Blog adds
B, P, D	$\varnothing$	$v(B) - 0 = .10$
B, D, P	$\varnothing$	.10
P, B, D	$\{P\}$	$v(BP) - v(P) = .20$
D, B, P	$\{D\}$	$v(BD) - v(D) = .15$
P, D, B	$\{P, D\}$	$v(BPD) - v(PD) = .15$
D, P, B	$\{P, D\}$	.15

Six "what does Blog add" numbers — one per order.

Average them = Shapley value

$$\phi_{\text{Blog}} = \frac{.10 + .10 + .20 + .15 + .15 + .15}{6} = \frac{0.85}{6} \approx 0.142$$

Same averaging for each channel:

$$\phi_{\text{Blog}} \approx .142, \qquad \phi_{\text{Pricing}} \approx .242, \qquad \phi_{\text{Demo}} \approx .317$$

$$\underbrace{.142 + .242 + .317}_{\text{the three Shapley values}} = \underbrace{0.70}_{v(\{B,P,D\})}$$

$v(\{B, P, D\})$ = conversion value of all three channels together.

Efficiency: the credits add up to the whole — **no credit lost or invented.**

The same thing, written compactly

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \left(v(S \cup \{i\}) - v(S)\right)$$

- The sum runs over every subset S that could be "already there" before i joins.
- The fraction is just the weight that makes *every join order equally likely*.
- Exactly the average we did by hand — now in one line.

Why Shapley? It's the *only* fair split

A wishlist any fair credit rule should satisfy:

- **Efficiency** — the credits add up to the total value.
- **Symmetry** — equal contributors get equal credit.
- **Null player** — a channel that adds nothing gets zero.
- **Additivity** — credit adds up across combined games.

Shapley is the **unique** rule satisfying all four.

Channels as players

$v(S)$ = conversion rate of journeys whose distinct channel set is **exactly** S.

Predict: which channel earns the most credit?

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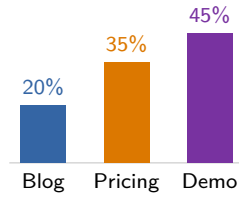


Shapley credit

First-touch crowned **Blog** (57%). Shapley says Blog earns just **20%** — **Demo** is the real driver.

The catch: 2^n coalitions

- With n channels there are 2^n subsets to value.
- 3 channels $\rightarrow$ 8 subsets (easy). 20 channels $\rightarrow$ over a million.
- Exact Shapley gets expensive fast $\Rightarrow$ in practice, *sample* orderings / approximate.

This cost — and the need for richer features — points to **Shapley on top of ML**.

Raw Shapley sees only presence

- It uses only *which* channels were present — a 0/1 flag per channel.
- Rare channel combinations have little or no data $\Rightarrow$ $v(S)$ is noisy or undefined.
- We want **richer features** and a way to **fill the gaps**.

Idea: let a *model* estimate $v(S)$ instead of raw counts.

Same math, a new value function

§4: empirical

$v(S)$ = conversion rate *counted from data* for set S

§5: model

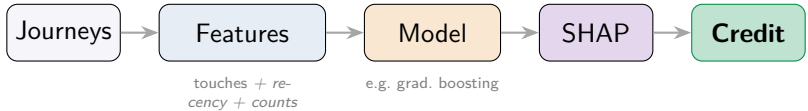
$v(S)$ = the model's *predicted* conversion prob. using only the features in S

Same Shapley formula — the value function goes from *counted-from-data* to *model-predicted*.

Why it helps: the model **interpolates** channel combinations the raw counts never saw.

This is **SHAP** — SHapley Additive exPlanations.

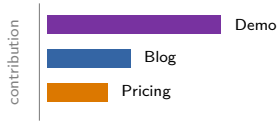
From journeys to SHAP credit



Features can now go *beyond* 0/1 — recency, frequency, dwell, and more.

From one prediction to channel-level credit

- SHAP explains *one user's* predicted probability — splitting it across that user's features.
- Average |SHAP| across all users ⇒ aggregate channel credit.



one journey's SHAP contributions to its predicted conversion probability

Everything so far treats a touch as 0/1

Remember the repeated Blog and Pricing visits we dropped on slide 1? That's **frequency** — gone.

Also invisible to these models:

- **Recency** — how recently each touch happened.
- **Repetition** — one visit vs ten.